Assessment Handbook

PS50008A Research Methods & Experimental Design

Term 2 · Spring 2026

Gordon Wright

Goldsmiths, University of London

How to Use This Handbook

This handbook is your single reference for all assessed work in PS50008A. It contains the brief, structure, marking information, and preparation advice for each of the four assessments.

Keep it handy throughout the term — everything you need is here.

Assessment Overview

Assessment	Weight	Deadline	Format
Research Plan	20%	Fri 27 Feb 2026, 12 noon	~1,000 word plan
MCQ Exam	20%	Wed 11 Mar 2026, in lecture	1 hr paper-based MCQ
EDS Assignment	20%	Fri 20 Mar 2026, 12 noon	1-2 hrs open-book results interpretation
Research Report	40%	1 May 2026, 12 noon	2,500 word APA report

All coursework submissions are via **Moodle** by **12 noon** on the stated deadline. The MCQ Exam takes place in your normal lecture slot (10-12 Wednesday RHBD 143).

How Assessments Fit Together

The four assessments are designed to build on each other progressively. The **Research Plan** gets you thinking about design; the **MCQ Exam** tests your conceptual understanding; the **EDS Assignment** checks you can interpret and report statistical output; and the **Research Report** brings everything together in a full write-up.

Your **group project**, which runs in labs in the second half of term, is the real experiment you design and run about which you write the Research Report — although this is group work, the Research Report is written and submitted individually.

Research Plan

Assessment 1 · 20%

At a Glance

Weight	20%
Deadline	Friday 27 February 2026, 12 noon
Submission	Via Moodle
Word count	Approximately 1,000 words
Format	Written plan (no data collection)

The Task

Describe how you would investigate **one** of the following research topics:

1. The effect of font difficulty on memory for information
2. The effect of hydration on cognitive performance
3. The effect of eye contact on perceived trustworthiness

This is a *plan*, not a real study. You do **not** collect data or invent results. This is SEPARATE from the Group Project.

Structure & Marking

Your plan should contain three sections:

a) Experimental Hypothesis

- State your hypothesis clearly
- Specify whether it is one-tailed or two-tailed
- Explain why you chose this direction

b) Method Section

- **Design**: Type of design and why it is appropriate
- **Participants**: Who, how many, how recruited
- **Materials/Stimuli**: What you would need
- **Procedure**: Step-by-step with appropriate controls

c) Draft Results Section

- A summary data table with appropriate headings only (do **not** invent data)
- State which statistical test should be used
- Justify your choice of test

Assessment Criteria

- Clarity and appropriateness of hypothesis
- Justification of design choices
- Feasibility and detail of procedure
- Accurate results/data layout
- Overall coherence and evidence of learning outcomes

Curriculum Connection

Week	Topic	How It Helps
11	The Big Picture	Overall design framework
12	Variables & Hypotheses	Writing hypotheses, identifying IVs/DVs
13	Experimental Design	Between/within designs, controls
15	Correlation	Understanding alternative designs

How to Prepare

1. **Choose your topic from the 3 options early** — they are all equally tricky
2. **Revisit lecture materials** from Weeks 11–13 on design and variables
3. **Practice coming up with hypotheses** — make them specific and testable
4. **Use the design** to guide the analytic approach
5. **Use your lab instructor well** — does the procedure make sense to them?

FAQs

Q: Do I need to find published references? A: References are not required throughout, but a single real reference is required to demonstrate your ability to find a relevant paper and present APA format references and in-text citation.

Q: Should I invent results? A: No. Your data table should have column headings (e.g., Condition, Mean, SD) but leave the cells empty or use placeholder labels. The point is to show you know *what* to measure. This is clearly demonstrated in the example and worked example.

Q: Can I choose a topic not on the list? A: No — you must choose one of the three listed topics.

Q: How detailed should the procedure be? A: Detailed enough that someone could replicate your study from the description alone.

Common Pitfalls

- Choosing a two-tailed hypothesis without justification (or vice versa)
- Describing a design that doesn't match the chosen statistical test
- Forgetting to mention controls (randomisation, counterbalancing, standardised instructions)
- Inventing data when the brief explicitly says not to
- Writing a methods section that is too vague to replicate

MCQ Exam

Assessment 2 · 20%

At a Glance

Weight	20%
Date	Wednesday 11 March 2026
Time	10:00-12:00 (lecture slot)
Duration	1 hour
Location	RHBD 143
Format	Paper-based, multiple choice

The Task

Answer a series of multiple-choice questions based on research scenarios (“Investigations”). Each Investigation describes a short study and asks you to identify the design, variables, hypothesis, appropriate test, and interpretation.

There will also be additional MCQs testing comprehension of broader course material up to that point.

Structure & Marking

Each Investigation typically has **5 or 6 questions**:

Question	What You Identify
1	Design (between / within / matched / correlational)
2	Independent variable (or recognise no IV)
3	Dependent variable
4	Appropriate hypothesis type
5	Appropriate statistical test
6	Interpretation of results

You need to know the following test selection framework:

Design	Parametric	Non-parametric
Between-participants	Independent *t*-test	
Within-participants	Paired *t*-test	
Correlational	Pearson’s *r*	Spearman’s rho

Assessment Criteria

- Correct identification of designs, variables, and hypotheses
- Appropriate test selection with understanding of the decision framework
- Accurate interpretation of statistical results

Curriculum Connection

Week	Topic	How It Helps
11	DataLab	Design framework overview; generating your dataset
12	Research Design Framework	IVs, DVs, design types, hypothesis types, test selection
13	Building Your Research	Operationalisation, design decisions, connecting design to analysis
14	Is Your Effect Real?	NHST, p-values, t-tests, confidence intervals, effect sizes
15	Correlation	Pearson's r, Spearman's rho, scatterplots, correlation vs causation
16	APA Report Writing	APA structure, Methods section, academic writing conventions
17	MCQ & EDS Preparation	Revision and integration of all concepts

How to Prepare

1. **Complete weekly self-assessment activities** — they mirror the exam format exactly
2. **Link designs to relevant tests**: Design → Variables → Hypothesis → Test → Interpret
3. **Don't memorise — understand** the logic behind test selection
4. **Work through scenarios** with a study buddy — explain *why* you'd choose each test
5. **Run the example MCQ in exam conditions** — you will be able to do well just by attending class and engaging with the material

FAQs

Q: Is it open-book? A: No. Closed-book, no notes, no collaboration.

Q: Do I need to calculate anything? A: No. No calculations required.

Q: What should I bring? A: A pen (not pencil). Nothing else is needed.

Q: How many Investigations will there be? A: The exact number may vary, but expect several scenarios plus additional standalone questions.

Q: Will there be negative marking? A: No. Unanswered or incorrect questions score zero — there is no penalty for guessing.

Common Pitfalls

- Confusing between-participants and within-participants designs
- Selecting the wrong t-test variety for the design

- Misidentifying the IV and DV (read the scenario carefully)
- Rushing through scenarios without reading all the details
- Forgetting that correlational designs do not have an IV

EDS Assignment

Assessment 3 · 20%

At a Glance

Weight	20%
Deadline	Friday 20 March 2026, 12 noon
Submission	Via Moodle
Format	1-2 hrs open book Worked data interpretation assignment
Pass mark	50%

The Task

You will receive a research scenario with **pre-generated results**. Your task is to interpret the output and demonstrate your understanding of the statistical analysis. No coding or hand calculations required.

Structure & Marking

Section A: Statistical Mechanics (50 marks)

Question	What You Do	Marks
1	State the researcher's hypothesis	5
2	Identify design type, IV, DV, data type	16–20
3	Interpret R descriptive output and report in APA format	10
4	Interpret R `t.test()` output and justify test choice	15
5	Write a full APA results paragraph from R output	5
6	Error detection — identify what's wrong with a given analysis	5

Section B: Methods and Theory (50 marks)

Question	What You Do
7	Suggest two methodological improvements
8	Discuss individual differences or extensions
9	Propose a more ecologically valid alternative

Assessment Criteria

- Correct identification of design, variables, and hypothesis
- Accurate reading and reporting of R output in APA format

- Sound justification for test selection (not just the right answer — explain *why*)
- Ability to detect errors and improvements in research
- Thoughtful, specific, and feasible methodological suggestions

Curriculum Connection

Week	Topic	How It Helps
14	*t*-tests in R	Reading `t.test()` output, APA reporting
15	Correlation in R	Interpreting correlation output
12–13	Design & Variables	Identifying design features from scenarios

How to Prepare

1. **Practice interpreting R output** from lab exercises (Weeks 12–16)
2. **Learn APA format** for reporting *t*-tests and correlations
3. **Understand test selection logic** — *why* a particular test, not just which one
4. **Practice figure interpretation** — look at visuals and use these to inform practice
5. **For Section B**, practice writing about methodology — think improvements, not just descriptions

FAQs

Q: Do I need to write R code or do calculations? A: No. All R output is provided. You need only interpret it.

Q: How long should Section B answers be? A: Questions 7 and 8 can be concise (a solid paragraph each). Question 9 requires a little bit more usually.

Q: Is there punitive marking? A: No. No punitive marking — if you get something wrong it doesn't reduce marks elsewhere. Think about all assessments as opportunities to win marks!

Common Pitfalls

- Reporting the wrong or imprecise numbers — you only have a couple of results to report, so check them carefully
- Forgetting to include all elements required for APA reporting of test results
- Thinking something is so obvious you don't need to include it. Just show your thinking
- In Section B, suggesting vague improvements ("use more participants") instead of specific ones
- Not answering all the questions

Research Report

Assessment 4 · 40%

At a Glance

Weight	40%
Deadline	1 May 2026, 12 noon
Submission	Via Moodle
Word count	2,500 words (±10%)
Format	Individual lab report in APA format

The Task

Write an individual research report based on your **group project data**. While your group collects data together, the report is written and submitted individually — each group member produces their own independent write-up.

Structure & Marking

Your report should follow standard APA format:

Section	What to Include
Title Page	Informative title, your name and student number
Abstract	~150 words summarising the study
Introduction	Background, rationale, hypothesis
Method	Design, Participants, Materials, Procedure
Results	Descriptives, inferential tests, effect sizes
Discussion	Interpretation, limitations, future directions
References	APA format

Statistical reporting in APA format:

- **t-test**: $t(\text{df}) = X.XX$, $p = .XXX$, $d = X.XX$
- Include 95% confidence intervals

Assessment Criteria

- Clarity and structure of writing
- Appropriate use of APA format throughout
- Quality and accuracy of statistical reporting
- Interpretation and discussion of findings
- Critical evaluation and suggestions for future research

Curriculum Connection

Week	Topic	How It Helps
12–15	Statistics & R labs	Results section: descriptives, CIs, tests

Week	Topic	How It Helps
16–17	Methods & Introduction	Writing these sections with proper structure
18–19	Introduction continued	Building a literature-based rationale
20	Discussion	Interpreting findings and evaluating limitations

How to Prepare

1. **Start writing early** — draft your Method section as soon as your group finalises the design
2. **Keep notes during data collection** — you'll need procedural details for the Method
3. **Use lab exercises as templates** for your Results section formatting
4. **Read published papers** in your topic area for Introduction and Discussion models
5. **Leave time for revision** — the Easter break is your writing window, not your starting point

FAQs

Q: Can group members have the same Introduction? A: No. Each report must be entirely your own work. You may cite the same sources, but the writing must be independent.

Q: What if our group project doesn't produce significant results? A: That's fine. Non-significant results are just as valid. Focus on accurate reporting and thoughtful discussion of what the findings mean (or don't mean).

Q: Do I need to include raw data? A: No. Report summary statistics and inferential test results. Raw data is not required. Gordon will supply your data output to you.

Q: How many references do I need? A: There is no set number, but a good Introduction typically draws on 3-5 relevant sources. If unsure - discuss this with your lab instructor or Gordon

Q: Does the word count include the reference list? A: No. The 2,500-word limit covers the main text (Abstract through Discussion including in-text citations!). Title page and Reference list are excluded.

Common Pitfalls

- Submitting a report that reads identically to a group member's (this is an academic integrity issue)
- Leaving the entire write-up to the Easter break without any earlier drafting
- Reporting results without effect sizes or confidence intervals
- Writing a Discussion that just repeats the Results instead of interpreting them
- Not linking back to the hypothesis in the Discussion
- Forgetting APA formatting conventions (italics for statistics, proper citation style)

Reference

Timelines, Policies & Support

Academic Integrity & AI Policy

All assessments must be your own work:

- **Research Plan**: Individual submission based on a set topic
- **MCQ Exam**: Closed-book, no collaboration
- **EDS Assignment**: Individual open-book, no collaboration on answers
- **Research Report**: Individual write-up, even though data comes from your group

AI Policy: Using AI tools to generate your submitted work is not permitted. You may use AI to help understand concepts, but all submitted work must be written by you.

If you are unsure whether something constitutes academic misconduct, ask before submitting — not after.

Support & Resources

Resource	Details
Lectures	Core concepts, worked examples, exam preparation
Labs	Hands-on practice and group discussions
Office hours	Individual or small-group questions (check Moodle for times)
Moodle VLE	Assessment briefs, practice materials, discussion forum
Course website	Lecture slides, lab activities, readings

Practice versions of all assessments will be provided with full support. We cannot give marks for practice attempts, but we will work through examples together in class.

End of Assessment Handbook · PS50008A · Spring 2026